PROJECT A60 · MODULE 55-708252

AI4Mobility

A multimodal assistive platform for parents supporting children with mobility difficulties. This deck covers the problem, the sources, the step splitter, the ethics and the honest register of what is not built.

PRESENTED BY

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AGENDA

What this deck covers

Background, method, evidence, and a register of what has not been built.

1. **Problem and background**
   Why parents need this, and where the ten exercise programmes come from.

2. **The step splitter**
   A seven-stage rule-based pipeline — not a language model — and how well it does.

3. **Evidence and ethics**
   Synthetic data, the results it supports, and why no children were recorded.

4. **The honest register**
   Built, partly built and not built, each with a stated next step.

THE TEAM

Team Deepminds — four roles, one platform

Four members, four defined responsibilities. This deck presents the research and documentation work.

Krushna Sai Teja Adhala

Data engineering and pipeline lead

Peer contribution split: 28%

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Model development and evaluation lead

Peer contribution split: 26%

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Pose estimation and visualisation lead

Peer contribution split: 24%

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THIS DECK

Research and documentation lead

Peer contribution split: 22%

01

Problem and background

Home physiotherapy arrives as continuous prose. A parent has to turn it into a sequence of correctly performed movements, unsupervised.

THE PROBLEM

From prose to practice, at home, every day

1. **Who this is for**
   Children with cerebral palsy, developmental delay, or recovering from injury, who are prescribed home physiotherapy.

2. **What the parent receives**
   Thorough but text-heavy guidance: continuous prose describing a whole set of exercises.

3. **What the parent has to do with it**
   Translate it into a sequence of correctly performed movements, unsupervised, often daily.

4. **Where the platform intervenes**
   At the translation step: steps, a picture per step, adjustable repetitions and holds, and read-aloud support.

5. **What it deliberately does not do**
   It does not diagnose, prescribe or replace a physiotherapist, and the source text stays visible throughout.

WHAT WAS BUILT

A single self-contained page, with no backend at all

No server, no accounts, no storage, no tracking. The privacy position is a consequence of the architecture.

What the platform offers

- Ten exercise programmes with the original source description always displayed.
- An editable description box: Convert to steps rebuilds the step list from any text.
- Personalisation by focus area, with difficulty scaling repetitions and holds.
- Read this step and Read full guide, using the browser speech interface.
- Accessibility: larger text, high contrast, reduced motion, keyboard navigation and ARIA labelling.

What that architecture means

- Nothing a family types or performs leaves their device.
- There is no account to create and no history to leak.
- The pose coach runs on-device, so no video is uploaded or recorded.
- The trade-off is real: no saved progress and no therapist dashboard.

PROVENANCE

Ten programmes, and where each one comes from

Five are sourced from published NHS leaflets. The rest are labelled as general practice.

Programme	Steps	Source
Lower Limb Exercises	6	Sheffield Children's NHS — General lower limb exercises
Core and Hip Stability	7	Sheffield Children's NHS — Core and hip stability exercises
Glute Strengthening	6	Sheffield Children's NHS — Glute strengthening
Calf Control	2	Sheffield Children's NHS — Calf control exercises
Wrist and Hand Exercises	7	Sheffield Children's NHS — Wrist exercises
Thread the Needle	7	Yoga Basics — Threading the Needle
Child's Pose	6	General practice (labelled as such)
Cat–Cow Stretch	4	General practice (labelled as such)
Standing Balance	4	General practice (labelled as such)
Seated Upper Body	5	General practice (labelled as such)

Source URLs are listed in the report and in the About and sources section of the platform itself.

02

The step splitter

A rule-based linguistic pipeline that turns a description into steps. No model weights, no network call.

The splitter is NOT a language model

The supervisor asked whether an LLM splits the exercises into steps. It does not, and the report says so explicitly.

What it is

- A rule-based linguistic splitter, implemented in JavaScript for the platform and mirrored in Python for the notebook.
- No model weights are loaded and no network call is made.
- Deterministic and inspectable: the same text always produces the same steps.
- Verified by 24 JavaScript splitter tests and 41 Python checks including full parity between the two implementations.

What the adapter seam is

- An LLMSplitterAdapter exists in both implementations and returns null.
- It is not connected to any model or provider.
- It is a seam for future work, not a hidden feature.
- The rule engine is the only thing that has ever produced a step in this project.

A seven-stage rule-based pipeline

- **1. Normalise, then 2. detect existing structure**
  Numbered or bulleted lists are preserved and never re-split, because the author already did the splitting.
- **3. Segment into sentences**
  With abbreviation and decimal protection, so "approx." and "1.5" do not end a sentence.
- **4. Split on connectives**
  On then, next, after that and semicolons — but only when both halves read as instructions.
- **5. Classify each fragment**
  As action, position, hold, reps, safety or context. Safety sentences are separated and never shown as steps to perform.
- **6. Extract holds and repetitions**
  Including word numerals such as "ten times" and ranges such as "5 to 10 seconds", which becomes an 8 second hold.
- **7. Score confidence per step**
  Low-confidence context fragments are dropped, and the mean confidence is reported in the interface.

FIGURE 11

How well the splitter does, stated plainly

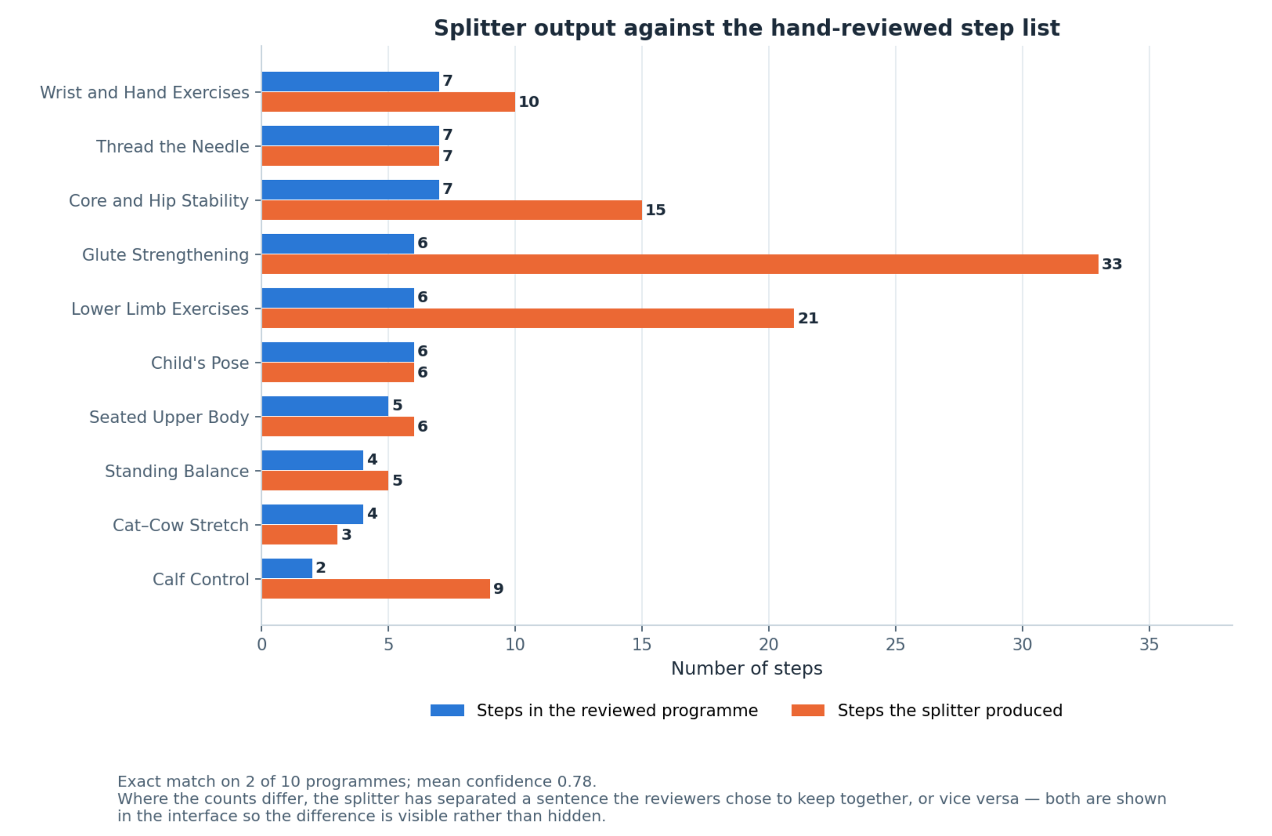


Figure 11 — splitter output against reviewed step counts for the ten programmes.

The honest result

- Exact match with the reviewed step list on 2 of 10 programmes.
- Mean confidence across all programmes is 0.78.
- Mismatches are on multi-exercise NHS leaflets, where one reviewed step covers a whole sub-exercise.
- The splitter produces more steps in those cases, not different instructions.

SHOWN, NOT HIDDEN

The interface displays the confidence and the original source text alongside the steps, so a parent can always see and check what was split.

FUTURE WORK

What connecting a language model would require

Four things, none of them optional, before the adapter could be switched on.

1. **A server-side proxy holding the API key**
   The platform is a static page, so a key cannot live in the client. That means a backend, which the project deliberately does not have today.

2. **A schema-constrained prompt and JSON validation**
   With the rule engine kept as the fallback whenever validation fails.

3. **Rate limiting**
   A public page calling a paid model needs a limit before it is exposed to anyone.

4. **A clinical safety review**
   A model can invent an instruction that was not in the source text. For physiotherapy instructions given to a child, that is the risk that matters.

STATUS

LLM-based step splitting is recorded as NOT built. The seam exists, the requirements are documented, and nothing in the platform pretends otherwise.

03

Evidence, ethics and honesty

What the numbers rest on, why no children were recorded, and a register of everything that is not built.

HONEST STATEMENT

The dataset is synthetic

Seeded, reproducible, and derived from published exercise descriptions — but not recorded video of children.

What it licenses us to claim

- The pipeline is complete, runs end to end, and reproduces exactly under random_state=42.
- The classes were constructed from published NHS descriptions, so they differ as the real exercises differ.
- The evaluation protocol is sound and would transfer unchanged to recorded data.
- The project can describe precisely what it would do with real data.

What it does not license

- No claim about accuracy on real children performing real physiotherapy.
- No clinical claim, and no claim of readiness for clinical use.
- No claim that the platform improves adherence, technique or outcomes.
- No comparison against published results on recorded datasets.

FIGURE 3

The evidence trail: every removal is counted

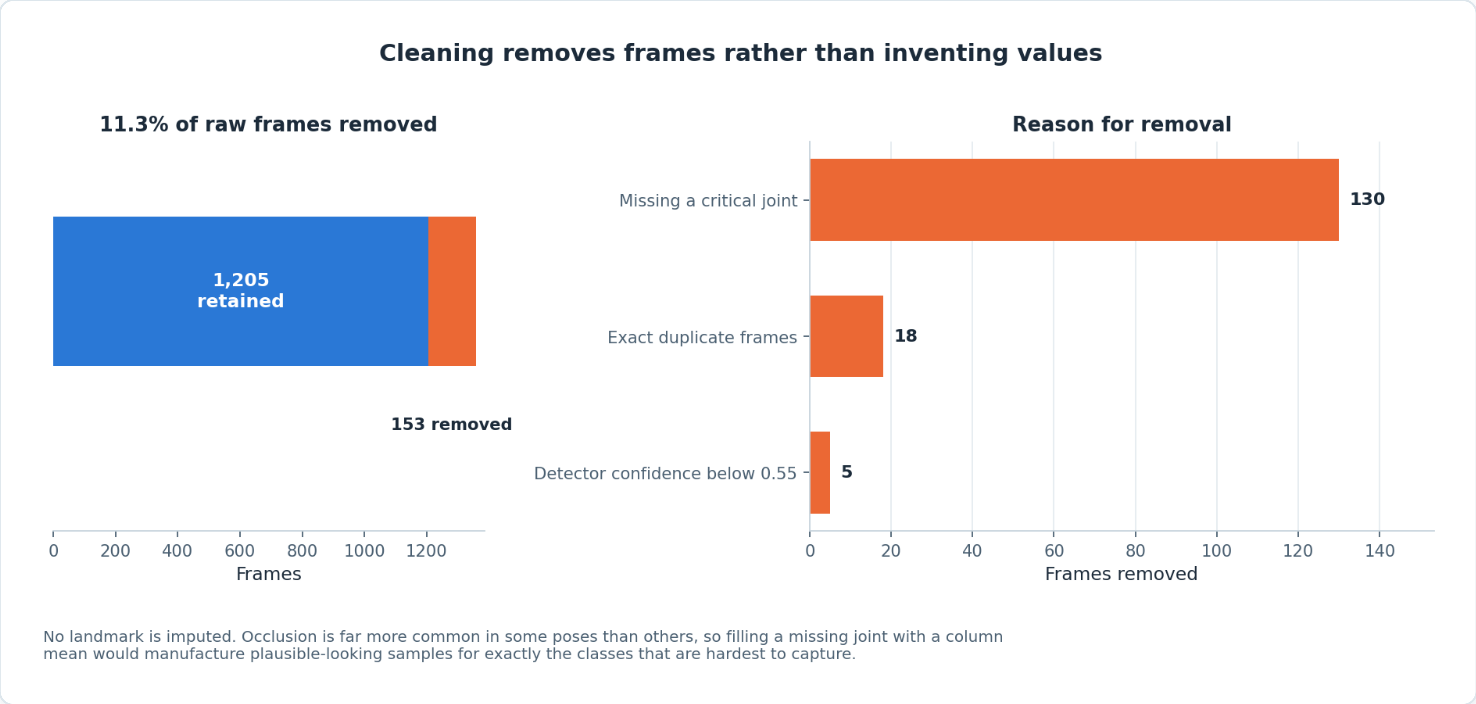


Figure 3 — 11.3% of raw frames removed, broken down by reason.

Why this belongs in a documentation deck

- 1,358 raw frames, 153 removed, 1,205 usable.
- Every removal is attributed to one of four stated rules.
- No landmark is imputed, because occlusion is pose-dependent and filling would manufacture samples for the hardest classes.
- A reader can reconstruct the dataset from the report without reading the code.

The classifier results, in full

Three model families, identical data and identical splits. Measured on synthetic data.

Model	Test accuracy	Test macro F1	CV macro F1 (5-fold, train only)	Train accuracy	Generalisation gap
Random Forest	**0.928**	**0.935**	**0.955 ± 0.017**	**0.965**	**+0.037**
Gradient Boosting	0.923	0.931	0.939 ± 0.016	1.000	+0.077
Neural Network (MLP)	0.923	0.932	0.939 ± 0.019	1.000	+0.077

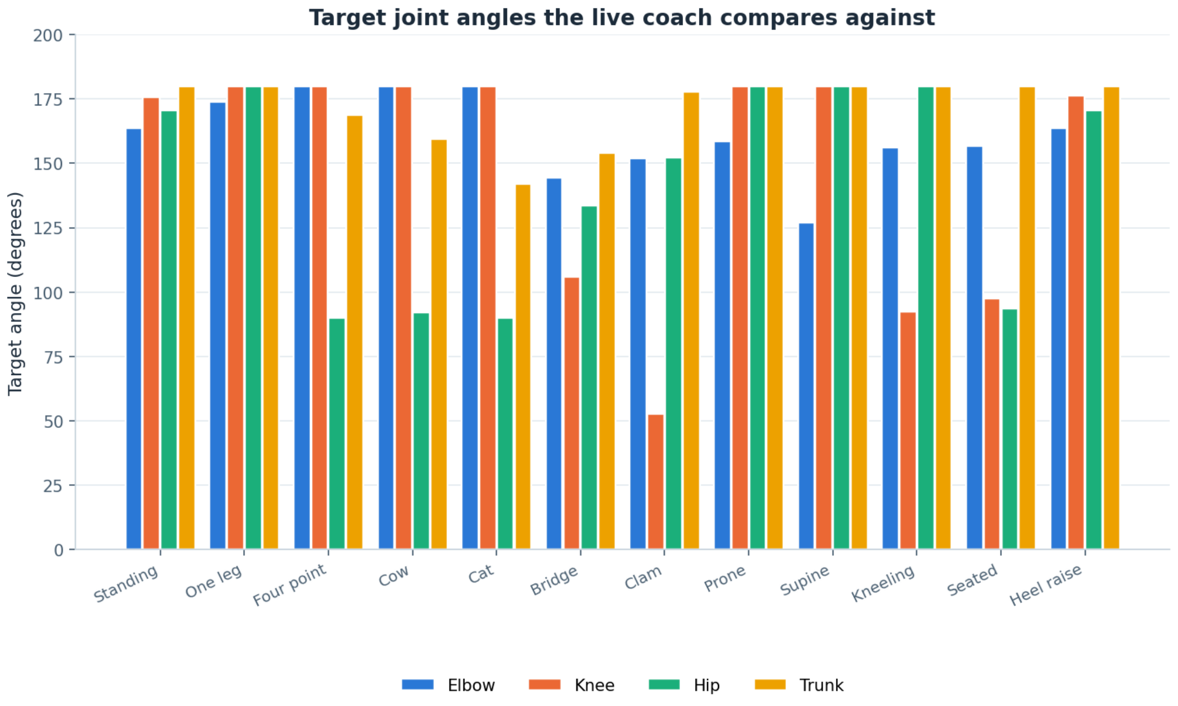
Cross-validation is five-fold on the training partition only. The test set was scored once, after all model selection was complete.

HOW TO QUOTE THESE NUMBERS

Random Forest is the best model at 0.928 accuracy and 0.935 macro F1, with the smallest generalisation gap at +0.037. Every quotation of these figures must carry the phrase "measured on synthetic data".

FIGURE 12

The pose coach, and the limit of its authority



Derived from the same hand-authored pose bank the illustrations are drawn from, so the coach checks the child against exactly the position the picture shows. Tolerances are 20–24° per joint and have not been clinically validated.

Figure 12 — target joint angles per pose family, derived from the hand-authored pose bank.

What is and is not established

- Targets are derived from the same hand-authored pose bank the illustrations are drawn from.
- Four joints are checked: elbow, knee, hip and trunk.
- Tolerances of 20 to 24 degrees are engineering judgement, not clinical thresholds.
- The coach runs on-device and degrades cleanly if there is no camera, no permission, or no model.

RECORDED AS NOT VALIDATED

Pose coach calibration is listed as NOT VALIDATED. No physiotherapist has reviewed the tolerances or the pose bank they come from.

Why no children were recorded

1. **Recording children requires approval this project does not have**
   Ethical approval, parental consent and safeguarding arrangements would all be needed before a single frame could be captured.

2. **A synthetic generator was the honest alternative**
   It allows the whole pipeline to be built and tested without asking a child to be filmed for a taught project.

3. **The platform stores nothing**
   No server, no accounts, no storage, no tracking. Nothing a family types or performs leaves the device.

4. **The pose coach never uploads a frame**
   Pose estimation runs on-device, and the video element is the only place the image exists.

5. **The interface stays inside its competence**
   Safety sentences are separated from steps, the source text is always shown, and the platform does not diagnose or prescribe.

What is not built, and what the next step would be

Requested by the supervisor, and reproduced in the report as its own section.

Item	Status	Next step
LLM-based step splitting	NOT built	Server-side key, schema validation, rate limiting, clinical safety review
Real recorded dataset	NOT obtained	Ethical approval, consent and safeguarding, then record a small consented set
Temporal / sequence modelling	NOT built	Add sliding-window features and re-run the same model comparison
Clinical validation	NOT done	Physiotherapist review of step lists, illustrations and coach tolerances
User evaluation	NOT done	Evaluate with families: is the exercise done, and done correctly
Accounts, history, therapist view	NOT built	Requires a backend, which brings data protection obligations with it
Generative illustrations	NOT built	46 hand-authored poses stand in for a generative vision model
Pose coach calibration	NOT validated	20 to 24 degree tolerances are engineering judgement, not clinical thresholds

What the evidence cannot support

- **No result here describes a real child.**
  The dataset is synthetic, so every metric is a property of a seeded generator.
- **No clinician has reviewed the content.**
  Step lists, illustrations and coach tolerances are unreviewed by a physiotherapist.
- **No family has used the platform.**
  Usability, comprehension and adherence are all untested.
- **The splitter agrees exactly with the reviewed lists on 2 of 10 programmes.**
  Mean confidence 0.78, with granularity differences on multi-exercise leaflets.
- **There is no backend, so there is no continuity.**
  No saved history, no progress over time, and no therapist view of what happened.

NEXT STEPS

Future work, in priority order

The five steps recorded in the report, in the order they should be taken.

1

1. Obtain ethical approval and record a consented dataset
Even 20 participants would let every number in this project be re-derived from reality.

2

2. Add temporal features and re-run the comparison
The confusion structure predicts exactly where the gain would appear.

3

3. Have a physiotherapist review the step lists and tolerances
The cheapest step that changes what the project is allowed to claim.

4

4. Connect a language model behind a validating proxy
Keeping the rule engine as the fallback, with schema validation and a safety review.

5

5. Evaluate with families
Does the platform change whether the exercises get done, and done correctly?

IN CLOSING

Say what was built, say what was not, say what is next

Sourced, not invented

Ten programmes, five from published Sheffield Children's NHS leaflets, the rest labelled as general practice.

Rule-based, and said so

A seven-stage deterministic splitter, no language model, an adapter seam that returns null and is documented as such.

Honest by construction

Synthetic data declared, 2 of 10 exact splitter matches published, eight unbuilt items each with a next step.

A project is more useful when its limits are written down than when they are discovered by the reader.

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